



CEBU INSTITUTE OF TECHNOLOGY
U N I V E R S I T Y

IT342-G6

System Integration and Architecture

System Design Document (SDD)

Project Title: Labang Online

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REVISION HISTORY TABLE

Version	Date	Author	Changes Made	Status
1.0	2/21/2026	Natividad, Lenon Lee	Initial draft	Draft
2.0	2/28/2026	Natividad, Lenon Lee	Final draft	Done

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EXECUTIVE SUMMARY

1.1 Project Overview

LabangOnline is a web-based barangay service system designed to modernize and digitize the delivery of services in Barangay Labangon, Cebu City. The system allows residents to request certificates, submit complaints, and track service requests online through a centralized platform for more efficient and transparent service delivery.

1.2 Objectives

1. Develop a fully functional barangay service system with user authentication and online service requests.
2. Implement a web-based platform for residents and barangay staff to manage services efficiently.
3. Create RESTful APIs to handle data communication between the system components.
4. Design a user-friendly and responsive interface for easy access on different devices.
5. Deploy the system in a secure environment for real-world barangay use.

1.3 Scope

Included Features:

- User registration and authentication (email/password)
- Online service requests (certificates, complaints, inquiries)
- Request tracking and status updates
- Admin panel for barangay staff to manage requests
- Dashboard with announcements and service overview
- Responsive web interface
- Supabase database with relational schema

Excluded Features:

- Online payment integration
- SMS or email notification system

- Mobile native application
- Social media login integration
- Push notifications
- Advanced analytics and reporting dashboard
- Dark/Light theme mode

1.0 INTRODUCTION

1.1 Purpose

This document serves as the design specification for the LabangOnline system. It outlines the system requirements, architecture, database design, and implementation plan to guide development and ensure efficient integration of all system components.

2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

2.1 Project Overview

Project Name: Labang Online

Domain: E-Governance / Public Service

Primary Users:

1. Barangay residents
2. Barangay administrators

Problem Statement: Residents experience long queues, delays, and limited access to barangay services due to manual and paper-based processes.

Solution: A digital barangay service platform that allows residents to request documents, submit complaints, and track services online for faster and more transparent service delivery.

2.2 Core User Journeys

Journey 1: First-time Customer Purchase

1. Users visit the Labang Online web or mobile application.
2. Click "Sign Up" and create an account.
3. Logs in to access the dashboard.
4. Browses available barangay services (e.g., certificate request, complaint, inquiry).
5. Selects a service and fills out the request form.

6. Submits the request and receives a confirmation.
7. Tracks the request status until it is completed.

Journey 2: Returning Customer Purchase

1. User logs in with existing credentials.
2. Navigates to the dashboard.
3. Selects a service (certificate, complaint, or inquiry).
4. Submits a new request or views previous requests.
5. Tracks request status updates.
6. Receives confirmation when the service is completed.

Journey 3: Administrator Product Management

1. Admin logs in with administrator credentials.
2. Navigates to the admin dashboard.
3. Views all service requests.
4. Approves, rejects, or updates request status.
5. Adds or edits barangay announcements and services.
6. Manages resident accounts and records.

2.3 Feature List (MoSCoW)

MUST HAVE

1. User authentication (register, login, logout)
2. Online service request (certificates, complaints, inquiries)
3. Request tracking and status updates
4. Admin panel for managing service requests
5. Dashboard with announcements and service overview)

SHOULD HAVE

1. User profile management
2. Request history for residents
3. Basic input validation and error messages
4. Responsive design for web and mobile

COULD HAVE

1. Document upload (supporting files for requests)
2. Basic reporting for admins
3. Search and filter for requests

WON'T HAVE

1. Online payment processing
2. SMS or email notifications
3. Social media login
4. Advanced analytics and dashboards
5. Multi-language support

2.4 Detailed Feature Specifications

Feature: User Authentication

- **Screens:** Registration, Login, Forgot Password
- **Fields:** Full Name, Email, Password, Confirm Password, Address
- **Validation:** Email format, password strength, uniqueness, required fields
- **API Endpoints:** POST /auth/register, POST /auth/login, POST /auth/logout
- **Security:** JWT tokens, password hashing with bcrypt, role-based access (Resident/Admin)

Feature: Service Management

- **Screens:** Service Listing, Service Request Form, Request Detail
- **Display:** List of barangay services (Certificate Request, Complaint Filing, Inquiry), status display
- **Search:** By service type or request status
- **API Endpoints:** GET /services, POST /requests, GET /requests/{id}, GET /requests/user
- **Admin Functions:** PUT /requests/{id}/status, GET /requests

Feature: Request Tracking

- **Screens:** My Requests
- **Functions:** View request history, track status (Pending, Approved, Rejected, Completed)
- **Persistence:** Database storage per user
- **API Endpoints:** GET /requests/user, GET /requests/{id}

Feature: Announcement Module

- **Screens:** Announcement List
- **Data Collected:** Announcement title, description, date posted
- **Process:** GET /announcements
- **API Endpoints:** POST /announcements, PUT /announcements/{id}, DELETE /announcements/{id}

Feature: Admin Panel

- **Screens:** Admin Dashboard, Request Management, User Management
- **Functions:** Approve/reject requests, update request status, manage announcements, view users
- **Access Control:** Admin role required
- **API Endpoints:** Admin-prefixed endpoints with role validation

2.5 Acceptance Criteria

AC-1: Successful User Registration

- Given I am a new resident
- When I enter a valid email, complete required personal details, and a strong password
- And confirm the password matches
- And click "Create Account"
- Then my account should be successfully created
- And I should be logged into the system
- And redirected to the resident dashboard

AC-2: Service Request Submission Flow

- Given I am logged in as a resident
- When I select a barangay service (e.g., certificate request, complaint, inquiry)
- And complete the required request form
- And submit the request
- Then I should see a confirmation message
- And the request status should be set to "Pending"
- And the request should appear in my request history
- And the request should be visible in the admin panel

AC-3: Admin Request Management

- Given I am logged in as an administrator
- When I add a new product with valid details
- And save the product
- Then the product should appear in customer product listing
- And be available for purchase

3.0 NON-FUNCTIONAL REQUIREMENTS

3.1 Performance Requirements

- API response time: ≤ 2 seconds for 95% of requests
- Web page load time: ≤ 3 seconds on broadband
- Mobile app cold start: ≤ 3 seconds
- Support 100 concurrent users
- Database queries complete within 500ms

3.2 Security Requirements

- HTTPS for all communications
- JWT token authentication
- Password hashing with bcrypt (salt rounds = 12)
- SQL injection prevention (parameterized queries / ORM)
- XSS protection through input validation and output encoding
- Rate limiting: 100 requests/minute per IP
- Admin endpoints require role verification

3.3 Compatibility Requirements

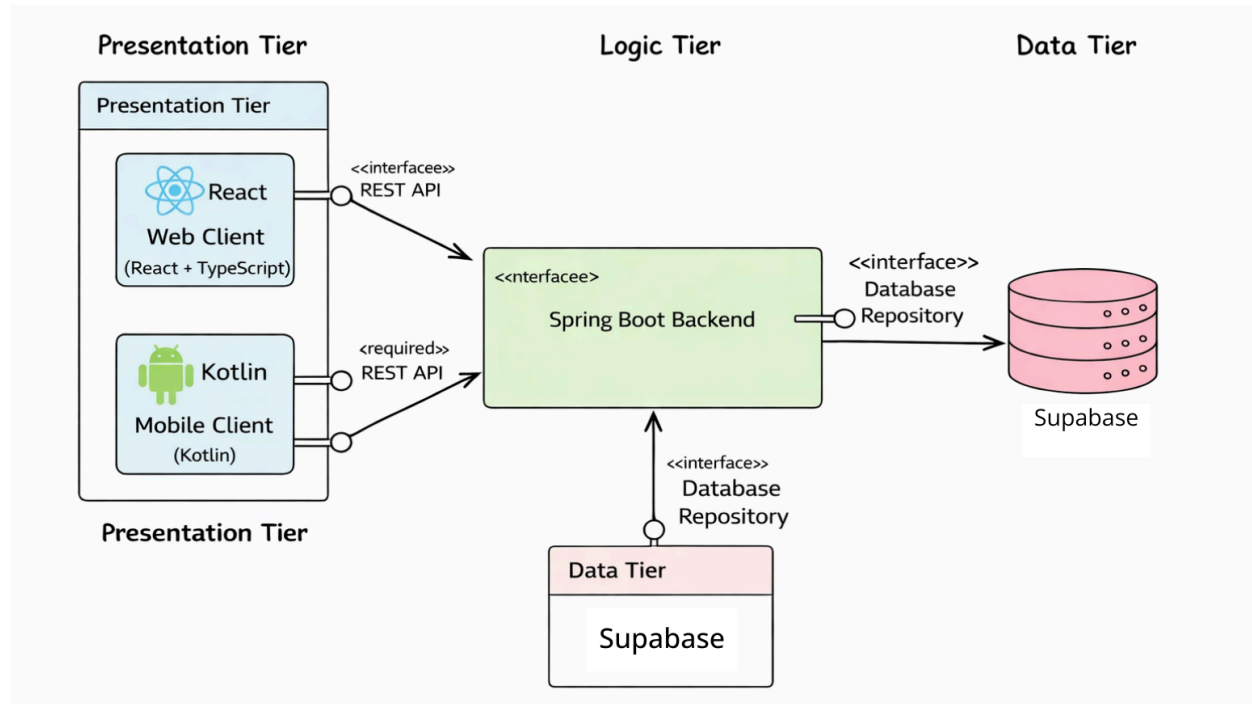
- **Web Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
- **Android:** API Level 24+ (Android 7.0+)
- **Screen Sizes:** Mobile (360px+), Tablet (768px+), Desktop (1024px+)
- **Operating Systems:** Windows 10+, macOS 10.15+, Linux Ubuntu 20.04+

3.4 Usability Requirements

- New residents should be able to complete a service request within 5 minutes
- WCAG 2.1 Level AA compliance for web accessibility
- Consistent navigation across web and mobile platforms
- Clear error messages with recovery options
- Touch targets minimum 44x44px on mobile
- Keyboard navigation support for accessibility

4.0 SYSTEM ARCHITECTURE

4.1 Component Diagram



Technology Stack:

- **Backend:** Java 17, Spring Boot 3.x, Spring Security, Spring Data JPA
- **Database:** Supabase
- **Web Frontend:** React 18, TypeScript, Tailwind CSS, Axios
- **Mobile:** Kotlin, Jetpack Compose, Retrofit, Room
- **Build Tools:** Maven (Backend), npm/yarn (Web), Gradle (Android)
- **Deployment:** Railway / Heroku (Backend), Vercel / Netlify (Web), Supabase (Database), APK Distribution (Mobile)

5.0 API CONTRACT & COMMUNICATION

5.1 API Standards

Base URL	https://[server_hostname]:[port]/api/v1
Format	JSON for all requests/responses
Authentication	Bearer token (JWT) in Authorization header
Response Structure	{ "success": boolean, "data": object null, "error": {

	<pre> "code": string, "message": string, "details": object null }, "timestamp": string } </pre>
--	---

5.2 Endpoint Specifications

Authentication Endpoints

User Registration

Description	User Registration
API URL	/auth/register
HTTP Request Method	POST
Format	JSON
Authentication	None
Request Payload	<pre> { "email": "<email>", "password": "<password>", "firstname": "<firstname>", "lastname": "<lastname>", "address": "<address>" } </pre>
Response Structure	<pre> { "success": true, "data": { "user": { "email": "<email>", "firstname": "<firstname>", "lastname": "<lastname>" }, "accessToken": "<token>", "refreshToken": "<token>" }, "error": null, "timestamp": "2026-01-28T10:30:00Z" } </pre>

User Login

Description	Authenticates resident or admin
API URL	/auth/login
HTTP Method	POST
Format	JSON

Authentication	None
Request Payload	<pre>{ "email": <email>, "password": <password>, }</pre>
Response Structure	<pre>{ "success": true, "data": { "user": { "email": "<email>", "firstname": "<firstname>", "lastname": "<lastname>", "role": "<RESIDENT ADMIN>" }, "accessToken": "<token>", "refreshToken": "<token>" }, "error": null, "timestamp": "2026-01-28T10:30:00Z" }</pre>

5.3 Error Handling

HTTP Status Codes

- 200 OK - Successful request
- 201 Created - Resource created
- 400 Bad Request - Invalid input
- 401 Unauthorized - Authentication required/failed
- 403 Forbidden - Insufficient permissions
- 404 Not Found - Resource doesn't exist
- 409 Conflict - Duplicate resource
- 500 Internal Server Error - Server error

Error Code Examples

Example 1 - Invalid Login	<pre>{ "success": false, "data": null, "error": { "code": "AUTH-001", "message": "Invalid credentials", "details": "Email or password is incorrect" }, }</pre>
---------------------------	--

	<pre>"timestamp": "2026-01-28T10:30:00Z" }</pre>
Example 2- Validation Error	<pre>{ "success": false, "data": null, "error": { "code": "VALID-001", "message": "Validation failed", "details": { "email": "Email is required", "password": "Must be at least 8 characters" } }, "timestamp": "2026-01-28T10:30:00Z" }</pre>

Common Error Codes

- AUTH-001: Invalid credentials
- AUTH-002: Token expired
- AUTH-003: Insufficient permissions
- VALID-001: Validation failed
- DB-001: Resource not found
- DB-002: Duplicate entry
- BUSINESS-001: Invalid request state (e.g., cannot update completed request)
- SYSTEM-001: Internal server error

6.0 DATABASE DESIGN

6.1 Entity Relationship Diagram

Detailed Relationships:

- **One-to-One:** User ↔ Requests (A resident can submit multiple service requests)
- **One-to-Many:** Admin → Announcements (An admin can post multiple announcements)
- **One-to-Many:** Requests → Service Types (Each request belongs to a service type, e.g., certificate, complaint, inquiry)

Key Tables:

1. **users** - Resident and admin accounts
2. **service_types** - Types of services offered by the barangay
3. **requests** - Residents' service requests

4. **announcements** - Barangay announcements
5. **orders** - Customer orders
6. **order_items** - Items in each order
7. **refresh_tokens** - JWT refresh tokens for authentication

Table Structure Summary:

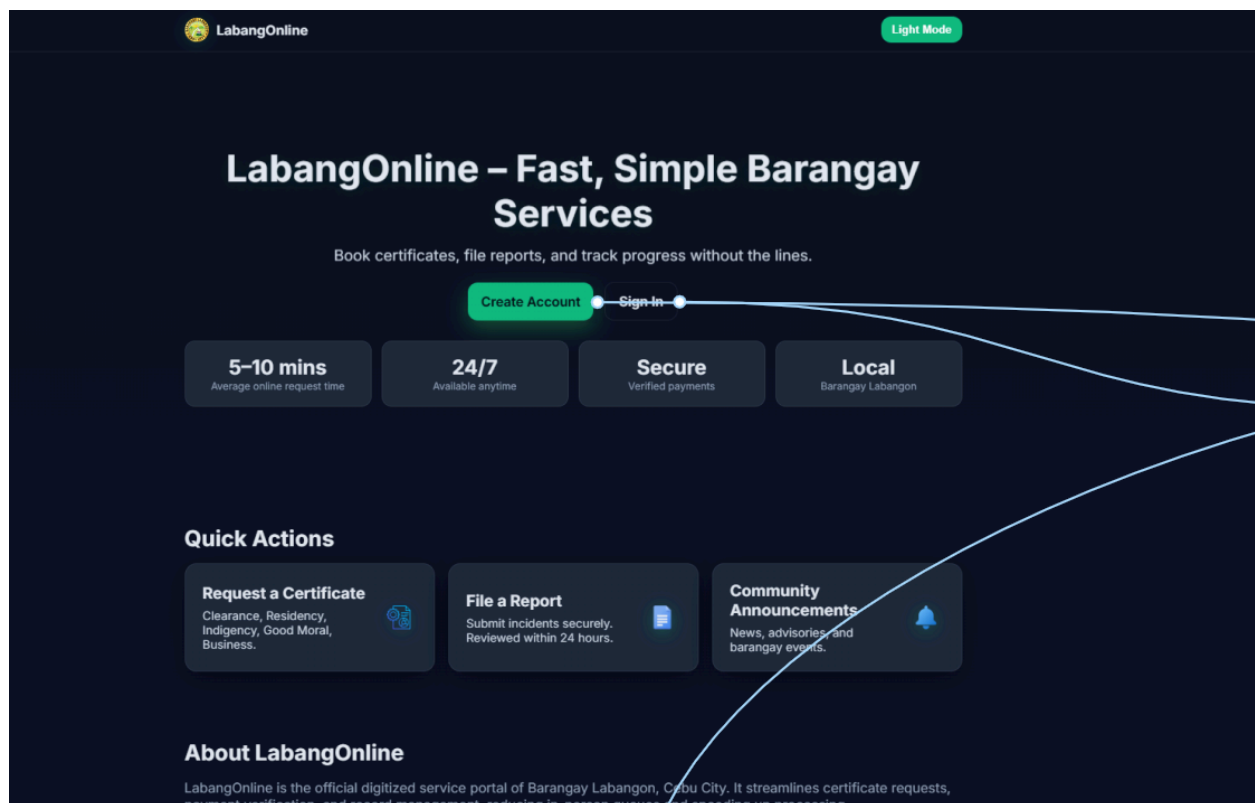
- **users:** id (PK), email, password_hash, full_name, role (RESIDENT / ADMIN), address, created_at
- **service_types:** id (PK), name, description, created_at
- **requests:** id (PK), user_id (FK → users.id), service_type_id (FK → service_types.id), request_details, status (Pending / Approved / Rejected / Completed), submitted_at, updated_at
- **announcements:** id (PK), admin_id (FK → users.id), title, description, posted_at
- **refresh_tokens:** id (PK) user_id (FK → users.id) token expires_at

7.0 UI/UX DESIGN

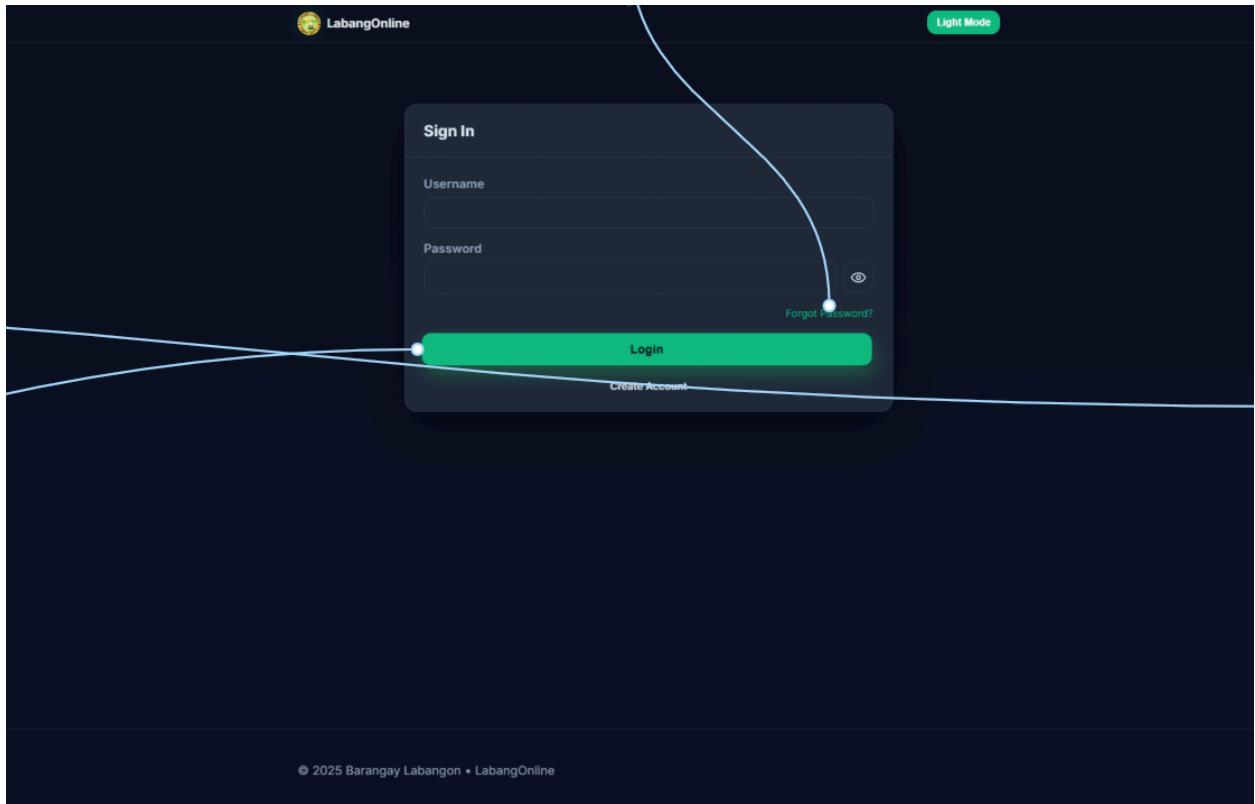
7.1 Web Application Wireframes

<https://www.figma.com/design/MrRGP1cM8dx76Eg65VwyRY/Untitled?node-id=0-1&t=mkwPWwgUDolrr8AY-1>

Landing Page



Login Page



The image shows the LabangOnline login page. At the top left is the LabangOnline logo, and at the top right is a 'Light Mode' toggle. The main content is a 'Sign In' form with fields for 'Username' and 'Password'. A 'Forgot Password?' link is located below the password field. A large green 'Login' button is centered below the fields. A 'Create Account' link is positioned below the 'Login' button. The footer contains the text '© 2025 Barangay Labangon • LabangOnline'.

LabangOnline

Light Mode

Sign In

Username

Password

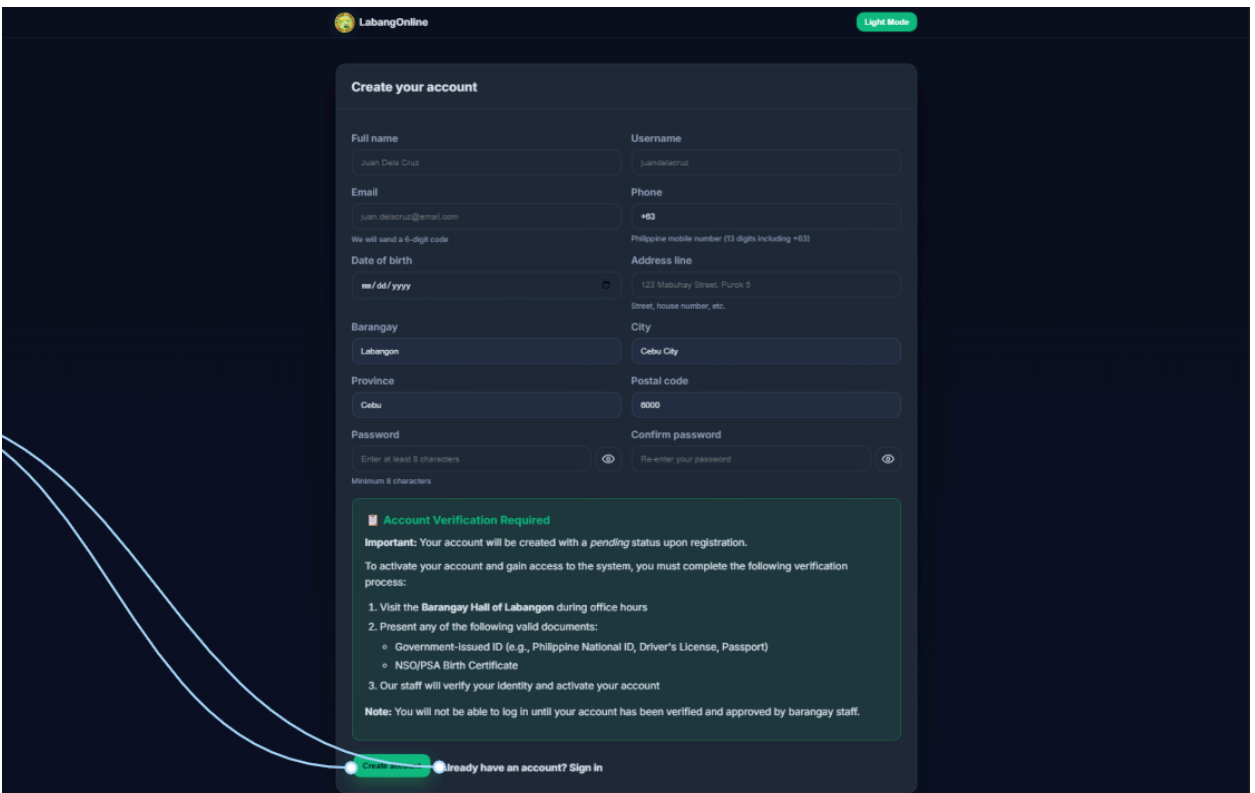
Forgot Password?

Login

Create Account

© 2025 Barangay Labangon • LabangOnline

Register Page



The image shows the LabangOnline registration page. At the top left is the LabangOnline logo, and at the top right is a 'Light Mode' toggle. The main content is a 'Create your account' form. It includes fields for 'Full name', 'Email', 'Date of birth', 'Barangay', 'Province', 'Password', 'Username', 'Phone', 'Address line', 'City', and 'Postal code'. A 'Create Account' button is at the bottom left, and a 'Already have an account? Sign In' link is at the bottom right. A green box contains 'Account Verification Required' information, including a list of steps to activate the account and a note about the pending status. The footer contains the text '© 2025 Barangay Labangon • LabangOnline'.

LabangOnline

Light Mode

Create your account

Full name

Username

Email

Phone

Date of birth

Barangay

Province

Password

Confirm password

Create Account

Already have an account? Sign In

Account Verification Required

Important: Your account will be created with a pending status upon registration.

To activate your account and gain access to the system, you must complete the following verification process:

1. Visit the Barangay Hall of Labangon during office hours
2. Present any of the following valid documents:
 - Government-issued ID (e.g., Philippine National ID, Driver's License, Passport)
 - NSO/PSA Birth Certificate
3. Our staff will verify your identity and activate your account

Note: You will not be able to log in until your account has been verified and approved by barangay staff.

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Forgot Password Page

LabangOnline Light Mode

Forgot Password

Enter your email address and we'll send you a 6-digit verification code via Gmail!

Email Address

you@example.com

Send Verification Code Back to Login

© 2025 Barangay Labangon • LabangOnline

Dashboard Page

LabangOnline Toggle Juan Dela Cruz LOGOUT

Announcements

Personal Info

Document Request

File Report

Juan Dela Cruz

RESIDENT

EDIT INFORMATION

USER INFORMATION

Full Name: Juan Dela Cruz

Username: juandelacruz

Email: juan.dela.cruz@email.com

Date of Birth: June 15, 1995

Civil Status: Single

Address: 123 Rizal St., Labangon, Cebu City, 6000

Resident Confirmed: ✓ Yes

CONTACT INFORMATION

Mobile: +63 917 123 4567

Email: juan.dela.cruz@email.com

RESIDENT ID

No Resident ID uploaded yet

Document Request

LabangOnline

Announcements

Personal Info

Document Request

File Report

Toggle

Juan Dela CruzLOGOUT

Request Certificate

Select the type of certificate you need and we'll guide you through the process

Important: Make sure your profile information is complete and accurate before requesting certificates. Processing time is typically 3-5 business days.

Barangay Clearance

Required for employment, business permits and other legal purposes within the Barangay.

P25.00

Certificate of Residency

Proof of residence in Barangay Labangon for government transactions and applications.

P30.00

Certificate of Indigency

For residents requiring financial assistance or applying for free medical services.

P20.00

Good Moral Character

Required for school applications, employment, and character reference purposes.

P40.00

Business Clearance

Required for starting or renewing a business within Barangay Labangon.

P100.00

General Requirements

Valid government-issued ID

Proof of residency in Barangay Labangon

Completed and verified user profile

Payment for processing fees (if applicable)

Additional documents may be required depending on certificate type

View Request Records

View Request Records

LabangOnline

Announcements

Personal Info

Document Request

File Report

Toggle

Juan Dela CruzLOGOUT

My Certificate Requests

Track the status of your certificate requests

+ New Request

TOTAL REQUESTS

7

PENDING PAYMENT

2

PAID

4

UNPAID

1

Certificate Type

Payment Status

Date From

Date To

All Types

All Status

Any Date

Any Date

REQUEST ID	CERTIFICATE TYPE	DATE REQUESTED	PURPOSE	PAYMENT	ACTION
REQ-001	Barangay Clearance	Feb 10, 2026	Employment	Paid	View
REQ-002	Certificate of Residency	Feb 14, 2026	Scholarship	Pending	View
REQ-003	Good Moral Character	Feb 18, 2026	School Application	Paid	View
REQ-004	Certificate of Indigency	Feb 20, 2026	Medical Assistance	Unpaid	View
REQ-005	Business Clearance	Feb 21, 2026	Business Permit	Pending	View

File Report

LabangOnline

Toggle

Juan Dela Cruz LOGOUT

File a Report

Report an incident to the Barangay Labangon

Note: Please provide as much detail as possible. False reports may be subject to legal action. Your report will be reviewed within 24-48 hours.

Incident Details

Incident Type: Noise Complaint

Date of Incident: 02/21/2026

Time of Incident: 02:30 PM

Location / Place: Purok 4, Near Labangon Market

Description of Incident: Loud music was being played at 2:30 PM causing disturbance to neighbors in the area.

Persons Involved (if known): Unknown

Additional Notes: Any other relevant information...

SUBMIT REPORT Cancel

View File Reports

LabangOnline

Toggle

Juan Dela Cruz LOGOUT

My File Reports

Track the status of your certificate requests

NEW REQUEST

TOTAL REQUESTS: 7

PENDING PAYMENT: 2

PAID: 4

UNPAID: 1

Certificate Type: All Types

Payment Status: All Status

Date From: Any Date

Date To: Any Date

REQUEST ID	CERTIFICATE TYPE	DATE REQUESTED	PURPOSE	PAYMENT	ACTION
REQ-001	Barangay Clearance	Feb 10, 2026	Employment	Paid	View
REQ-002	Certificate of Residency	Feb 14, 2026	Scholarship	Pending	View
REQ-003	Good Moral Character	Feb 18, 2026	School Application	Paid	View
REQ-004	Certificate of Indigency	Feb 20, 2026	Medical Assistance	Unpaid	View
REQ-005	Business Clearance	Feb 21, 2026	Business Permit	Pending	View

Announcements

LabangOnline

Announcements

Personal Info

Document Request

File Report

Toggle

Juan Dela Cruz

Logout

Barangay Announcements

Stay updated with the latest news and events from Barangay Labangon

All

General

Events

Alerts

Maintenance

GENERAL

Barangay Health Fair — Free Medical Check-up
February 24, 2026 • 9:00 AM
The Barangay Labangon Health Center invites all residents to join our free medical check-up event. Services include blood pressure monitoring, blood sugar testing, dental check-up, and eye examination...
Posted by: Kapitan Pedro Santos
[Read more →](#)

EVENT

Barangay Fiesta Celebration 2026
March 5, 2026 • 6:00 AM
We are pleased to announce the annual Barangay Labangon Fiesta! Join us for a week of festivities celebrating the feast of our patron saint. Activities include street dancing, sports tournaments, and nightly entertainment...
Posted by: Administrator
[Read more →](#)

ALERT

Water Service Interruption Notice
February 22, 2026 • 8:00 AM
Please be informed that MCWD will conduct scheduled maintenance work resulting in water service interruption on February 25, 2026 from 8:00 AM to 5:00 PM affecting Purok 1 to Purok 6 of Barangay Labangon...
Posted by: Kapitan Pedro Santos
[Read more →](#)

MAINTENANCE

Admin Dashboard

Admin Panel

Pedro Santos

Dashboard

Users

Certificates

Reports

Announcements

User View

Logout

Dashboard Overview

Welcome back, Pedro Santos. Here's what's happening with your barangay.

TOTAL USERS
248

TOTAL ADMIN
4

VERIFIED USERS
201

PENDING VERIFICATION
47

TOTAL CERTIFICATES
312

PENDING PAYMENTS
28

PAID CERTIFICATES
284

TOTAL REPORTS
63

PENDING REPORTS
12

RESOLVED REPORTS
51

Recent Certificate Requests

REQ-012 — Barangay Clearance
Maria Santos • Feb 21, 2026
Pending

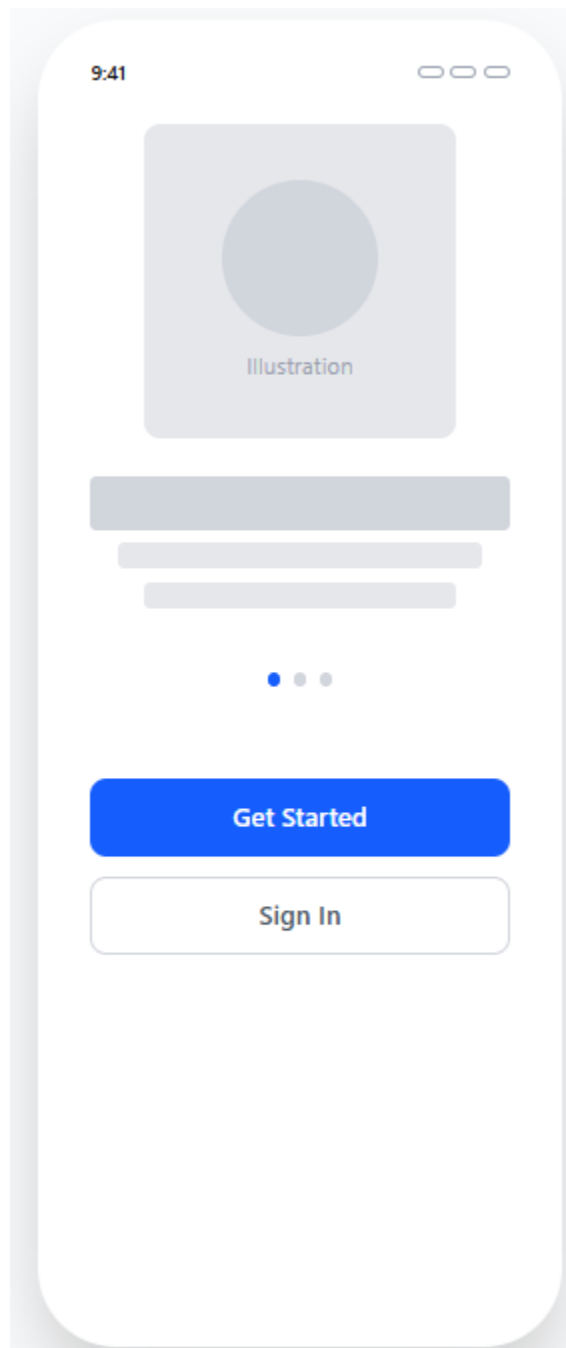
REQ-011 — Certificate of Residency
Jose Reyes • Feb 20, 2026
Paid

REQ-010 — Good Moral Character
Ana Lopez • Feb 19, 2026
Paid

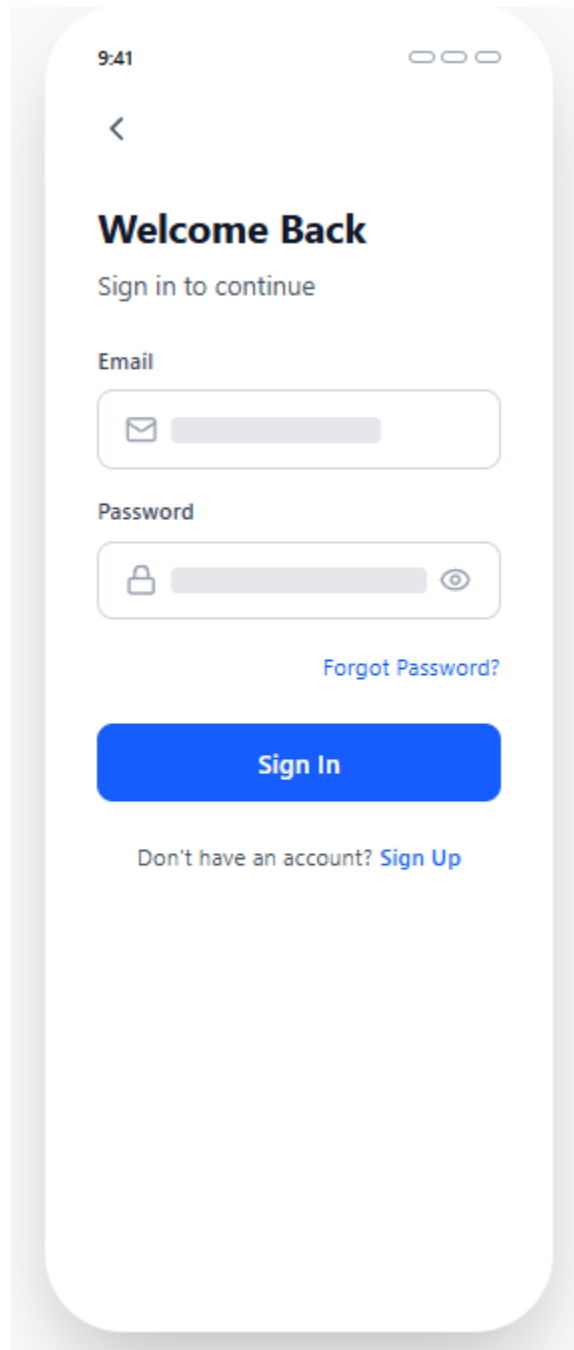
[View All →](#)

7.2 Mobile Application Wireframes

Landing



Login



A mobile app login screen mockup. At the top, the status bar shows the time 9:41 and three empty notification slots. Below the status bar is a back arrow icon. The main heading is "Welcome Back" in bold black text, followed by the subtitle "Sign in to continue". There are two input fields: "Email" with an envelope icon and "Password" with a lock icon and a toggle eye icon. A blue "Sign In" button is centered below the fields. To the right of the password field is a link "Forgot Password?". At the bottom, there is a link "Don't have an account? Sign Up".

9:41

<

Welcome Back

Sign in to continue

Email

✉

Password

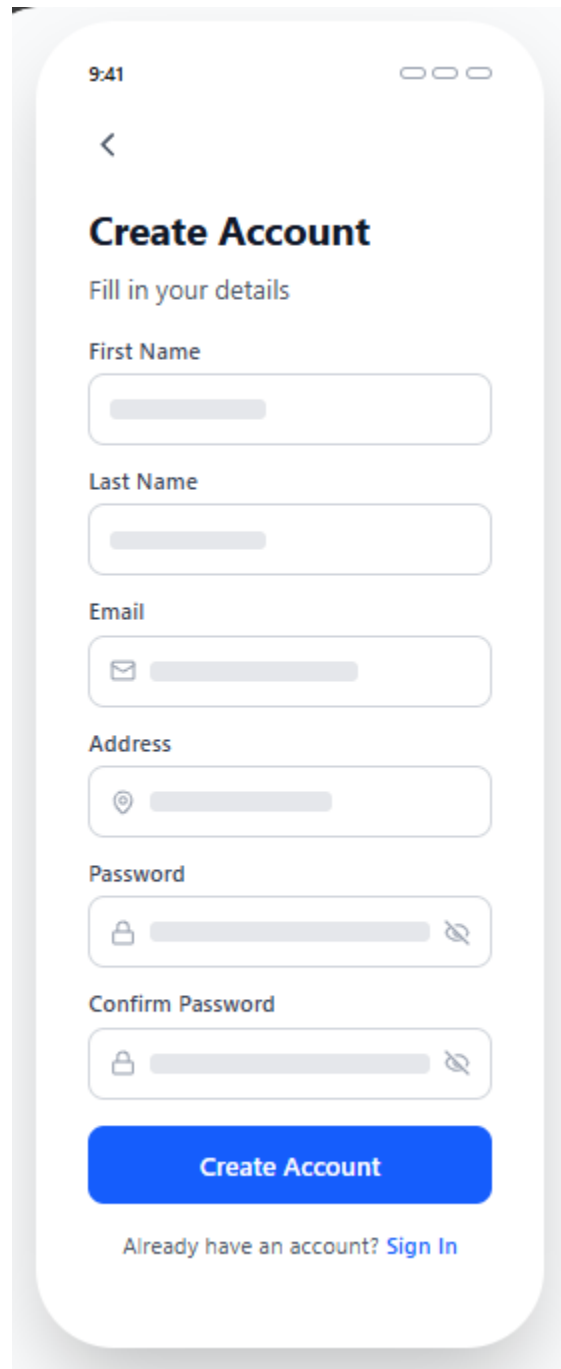
🔒

[Forgot Password?](#)

[Sign In](#)

Don't have an account? [Sign Up](#)

Register

A mobile application registration screen with a white background and rounded corners. At the top, the status bar shows the time 9:41 and three empty notification slots. Below the status bar is a back arrow icon. The main heading is "Create Account" in bold black text, followed by the instruction "Fill in your details". The form consists of several input fields: "First Name" and "Last Name" are simple text boxes; "Email" includes an envelope icon on the left; "Address" includes a location pin icon on the left; "Password" and "Confirm Password" include a lock icon on the left and an eye icon on the right to toggle visibility. A prominent blue button with the text "Create Account" is positioned below the password fields. At the bottom, there is a link that says "Already have an account? Sign In".

9:41

<

Create Account

Fill in your details

First Name

Last Name

Email

Address

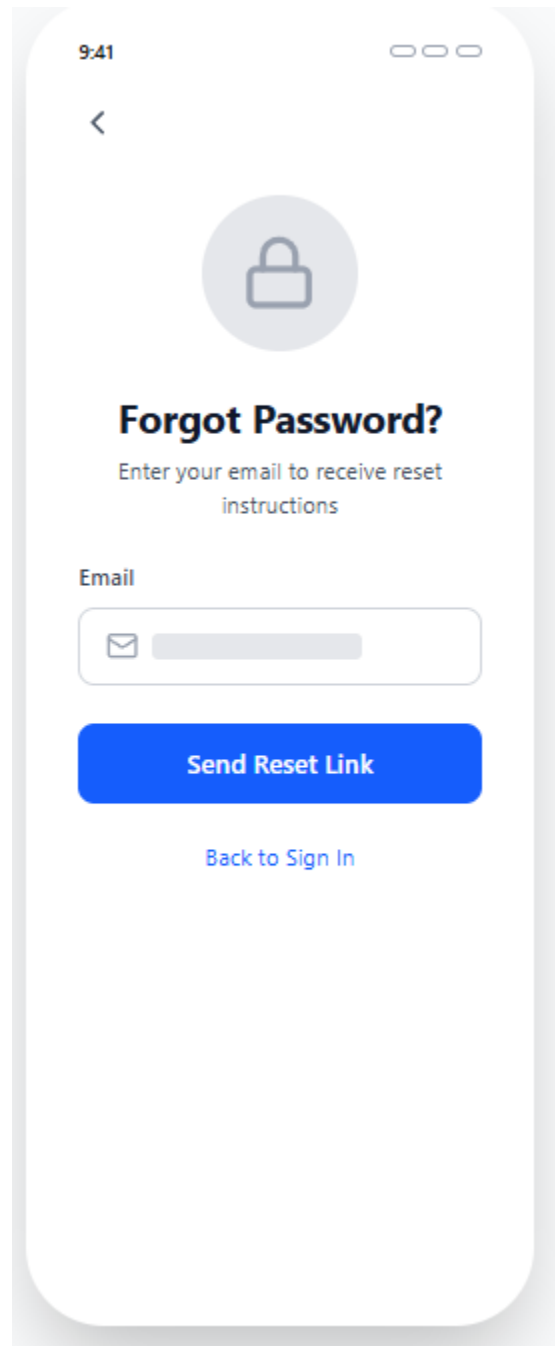
Password

Confirm Password

Create Account

Already have an account? [Sign In](#)

Forgot Password

A mobile application screen for password recovery. At the top, the status bar shows the time 9:41 and three navigation icons. Below the status bar is a back arrow icon. In the center, there is a circular icon containing a padlock. Below this icon, the text "Forgot Password?" is displayed in a bold font, followed by the instruction "Enter your email to receive reset instructions" in a smaller font. Below the instruction is a text input field with the label "Email" and an envelope icon. At the bottom of the input field is a grey bar. Below the input field is a blue button with the text "Send Reset Link". At the very bottom of the screen is a link that says "Back to Sign In".

9:41

<



Forgot Password?

Enter your email to receive reset instructions

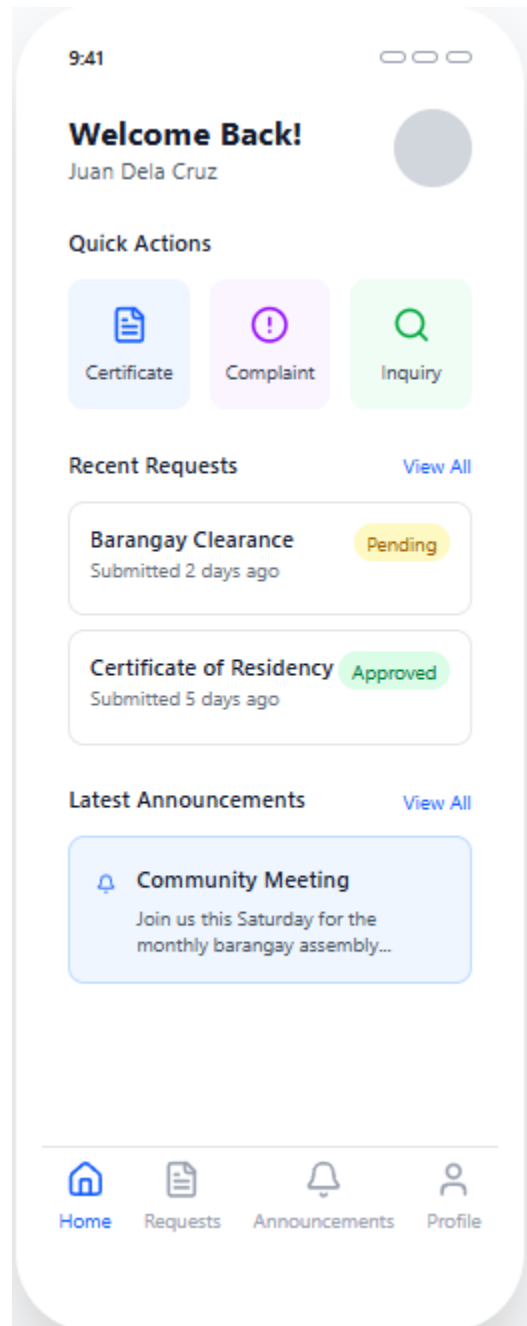
Email



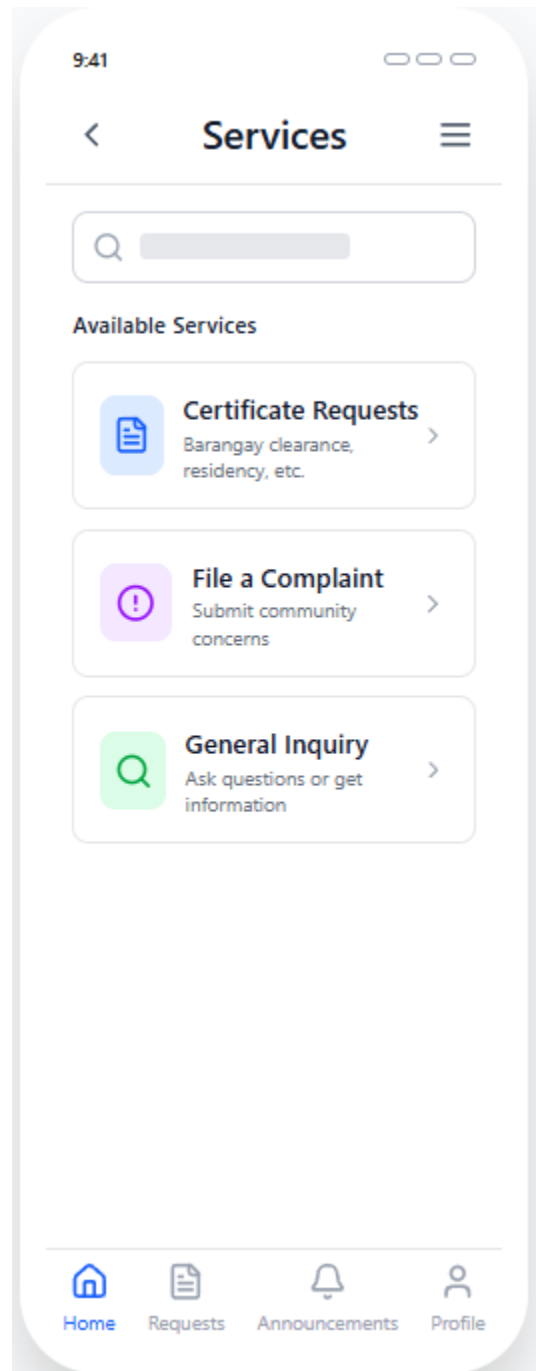
[Send Reset Link](#)

[Back to Sign In](#)

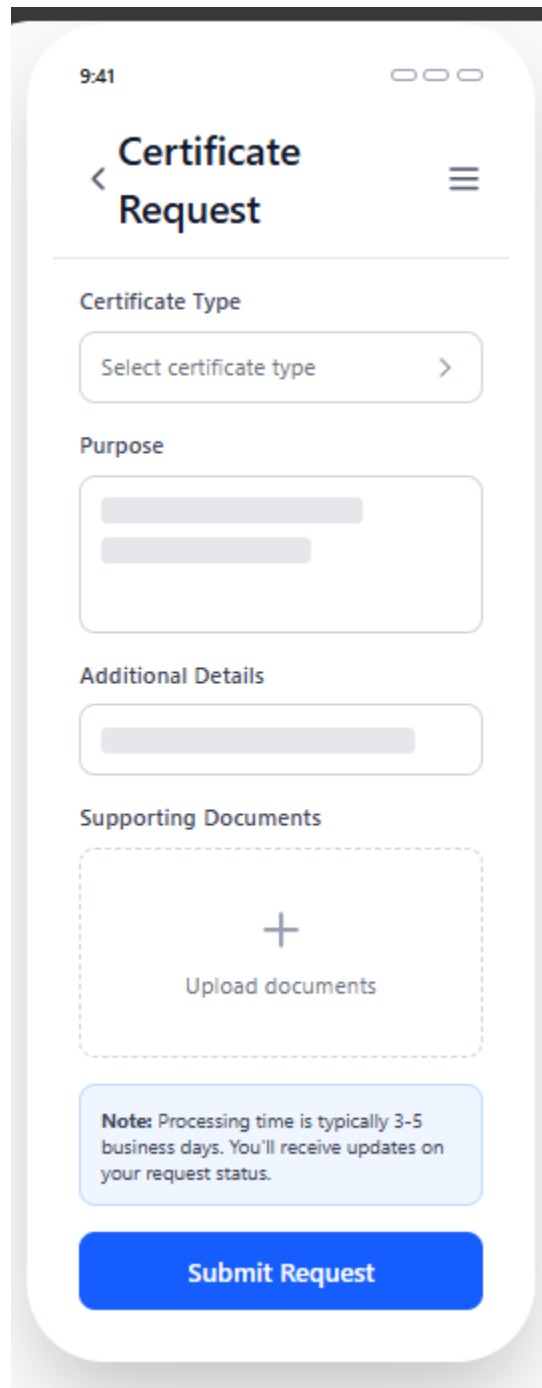
Dashboard



Services



Document Request



A mobile application interface for submitting a certificate request. The screen is titled "Certificate Request" with a back arrow on the left and a menu icon on the right. The form is divided into several sections: "Certificate Type" with a dropdown menu, "Purpose" with two text input fields, "Additional Details" with a single text input field, and "Supporting Documents" with a dashed box containing a plus icon and the text "Upload documents". A blue note box at the bottom states: "Note: Processing time is typically 3-5 business days. You'll receive updates on your request status." Below the note is a large blue "Submit Request" button.

9:41

< Certificate Request

Certificate Type

Select certificate type >

Purpose

Additional Details

Supporting Documents

+
Upload documents

Note: Processing time is typically 3-5 business days. You'll receive updates on your request status.

Submit Request

File Complaint

The image shows a mobile application interface for filing a complaint. At the top, the status bar displays the time 9:41 and three empty notification slots. Below this is a header bar with a back arrow, the title 'File Complaint', and a hamburger menu icon. The form consists of several sections: 'Complaint Type' with a dropdown menu labeled 'Select complaint type'; 'Subject' with a single-line text input; 'Description' with a multi-line text area; 'Location' with a location pin icon and a text input; 'Evidence (Photos/Videos)' with a dashed border box containing a plus icon and the text 'Add photos or videos'; a confirmation checkbox labeled 'I confirm that the information provided is true and accurate'; and a large blue 'Submit Complaint' button at the bottom.

9:41

< File Complaint ≡

Complaint Type

Select complaint type >

Subject

Description

Location

Evidence (Photos/Videos)

+
Add photos or videos

☐ I confirm that the information provided is true and accurate

Submit Complaint

Mobile-Specific Features:

- Touch-optimized buttons (min 44x44px)
- Gesture support (swipe, pull-to-refresh)

- Offline caching for product images
- Bottom navigation for main actions
- Simplified forms for mobile input

Design System:

- **Colors:** TBA
- **Typography:** Inter font family, responsive sizing
- **Spacing:** 8px grid system
- **Components:** Consistent buttons, inputs, cards, modals
- **Responsive:** Mobile-first approach, breakpoints at 640px, 768px, 1024px

8.0 PLAN

8.1 Project Timeline

Phase 1: Planning & Design (Week 1-2)

Week 1: Requirements & Architecture

Day 1-2: Project setup and documentation

Day 3-4: Complete Functional and Non-Functional Requirements

Day 5-7: System architecture design (Three-tier architecture)

Week 2: Detailed Design

Day 1-2: API specification (Authentication, Requests, Announcements)

Day 3-4: Database design (ERD and schema)

Day 5-6: UI/UX wireframes (Web + Mobile)

Day 7: Implementation plan finalization

Phase 2: Backend Development (Week 3-4)

Week 3: Foundation

Day 1: Spring Boot setup with dependencies

Day 2: Supabase PostgreSQL configuration and entity setup

Day 3: JWT authentication implementation

Day 4: User management endpoints

Day 5: Service type and request entity setup

Week 4: Core Features

Day 1: Service request submission API

Day 2: Request status update logic (Admin)

Day 3: Announcement management API

Day 4: Error handling and validation

Day 5: API documentation and testing

Phase 3: Web Application (Week 5-6)

Week 5: Frontend Foundation

Day 1: React setup with TypeScript

Day 2: Authentication pages (login, register)

Day 3: Servicelisting page

Day 4: Service detail page

Day 5: Request tracking page

Week 6: Complete Web Features

Day 1: Admin dashboard

Day 2: Request management interface

Day 3: Announcement management

Day 4: Responsive design polish

Day 5: API integration and testing

Phase 4: Mobile Application (Week 7-8)

Week 7: Android Foundation

Day 1: Android Studio setup and project structure

Day 2: Authentication screens

Day 3: Service browsing

Day 4: Service request submission

Day 5: API service layer (Retrofit integration)

Week 8: Complete Mobile App

Day 1: Request tracking screen

Day 2: Admin features (if included in mobile scope)

Day 3: UI polish and usability improvements

Day 4: Testing on emulator/device

Day 5: APK generation and documentation

Phase 5: Integration & Deployment (Week 9-10)

Week 9: Integration Testing

Day 1: End-to-end testing (Web + Mobile + Backend)

Day 2: Bug fixes and optimization

Day 3: Security review

Day 4: Performance testing

Day 5: Documentation updates

Week 10: Deployment

Day 1: Backend deployment (Railway/Heroku/Render)

Day 2: Web app deployment (Vercel/Netlify)

Day 3: Supabase production configuration

Day 4: Final system testing

Day 5: Project submission

Milestones:

- **M1 (End Week 2):** All design documents complete
- **M2 (End Week 4):** Backend API fully functional
- **M3 (End Week 6):** Web application complete
- **M4 (End Week 8):** Mobile application complete
- **M5 (End Week 10):** Full system deployed and integrated

Critical Path:

1. Authentication system (Week 3)
2. Service request API (Week 4)
3. Request tracking functionality (Week 6)
4. Mobile API integration (Week 7–8)
5. Cross-platform testing (Week 9)

Risk Mitigation:

- Start with the simplest working version of authentication and service request flow
- Test API integration early between backend and web
- Maintain version control backups
- Focus on core barangay service functionality before enhancements